ENGR 225: Mechanics 1

Homework Coversheet

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DATE: 10/8/2026

ASSIGNMENT: Homework 4

Level of involvement of AI, LLMs, or related tools in the completion of this assignment:

☐ None

■ Minor

☐ Major

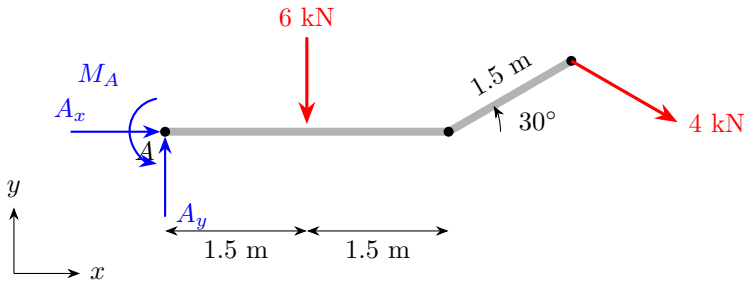
☐ Extensive

Mark the box that applies (e.g., replace ☐ with ■, or type an X beside it, before compiling/printing). Your disclosure does not affect your grade – it keeps you accountable for the work you are doing.

Problem 1

Given / Find: Cantilever beam with a 6 kN load at mid-span and a 4 kN force at 30° below horizontal on the end of the inclined segment. Find A_x, A_y, M_A.

Sketch:



Equations & Solution: ($+x$ right, $+y$ up, +ccw)

$$\sum F_x = 0: \quad A_x + 4\cos 30^\circ = 0 \;\Rightarrow\; A_x = -3.46\,\text{kN}$$

$$\sum F_y = 0: \quad A_y - 6 - 4\sin 30^\circ = 0 \;\Rightarrow\; A_y = 8.00\,\text{kN}$$

The 4 kN force acts at $x = 3 + 1.5\cos 30^\circ = 4.299\,\text{m}$, $y = 1.5\sin 30^\circ = 0.750\,\text{m}$. Moments about A:

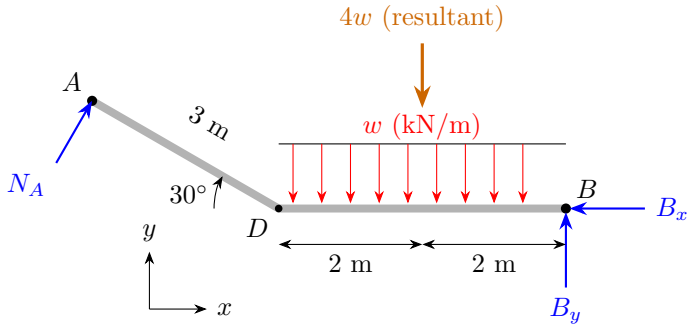
$$\sum M_A = 0: \quad M_A - (6)(1.5) - (4\sin 30^\circ)(4.299) - (4\cos 30^\circ)(0.750) = 0 \;\Rightarrow\; M_A = 20.2\,\text{kN m}$$

Answer
$A_x = -3.46\,\text{kN}$ (pointing left), $A_y = 8.00\,\text{kN}$ (up), $M_A = 20.2\,\text{kN m}$ (counterclockwise)

Problem 2

Given / Find: Roller at A (on the 30° incline) and pin at B support a 4 m span under uniform load w. Roller limit is 4 kN, pin limit is 8 kN. Find the largest w.

Sketch:



Equations & Solution: The roller force N_A is perpendicular to the incline. The distributed load becomes $4w$ at 2 m from B. The perpendicular distance from B to the line of action of N_A is $3 + 4\cos 30° = 6.464\,\text{m}$.

$$\sum M_B = 0: \quad 4w(2) - N_A(6.464) = 0 \;\Rightarrow\; N_A = 1.238\,w$$

$$\sum F_x = 0: \quad B_x = N_A \sin 30° = 0.619\,w$$

$$\sum F_y = 0: \quad B_y = 4w - N_A \cos 30° = 2.928\,w$$

$$B = \sqrt{B_x^2 + B_y^2} = 2.993\,w$$

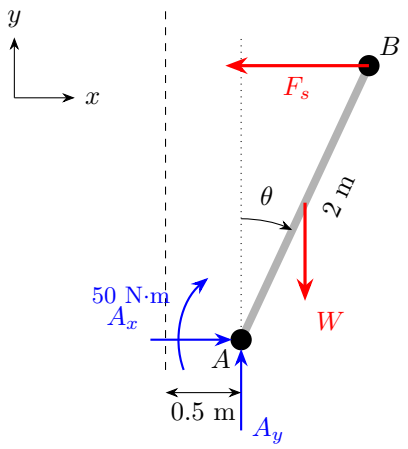
Roller: $1.238w \le 4 \Rightarrow w \le 3.23\,\text{kN/m}$. Pin: $2.993w \le 8 \Rightarrow w \le 2.67\,\text{kN/m}$. The pin governs.

Answer
$w_{\max} = 2.67\,\text{kN/m}$ (pin B governs; roller carries $N_A = 3.31\,\text{kN} < 4\,\text{kN}$)

Problem 3

Given / Find: A 10 kg, 2 m uniform rod is pinned at A and held by a horizontal spring ($k = 60\,\text{N/m}$, unstretched at $\theta = 0°$) at B, with a $50\,\text{N m}$ couple at A. Find the smallest θ for equilibrium.

Sketch:



Equations & Solution: Spring force (stretch is $2\sin\theta$) and weight:

$$F_s = k(2\sin\theta) = 120\sin\theta\ \text{N}, \qquad W = (10)(9.81) = 98.1\,\text{N}$$

Moments about A (spring force is ccw at height $2\cos\theta$; weight and couple are cw):

$$\sum M_A = 0: \quad F_s(2\cos\theta) - W\sin\theta - 50 = 0$$
$$120\sin 2\theta - 98.1\sin\theta - 50 = 0$$

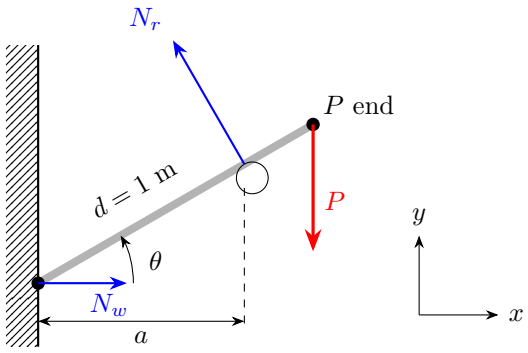
Solved numerically, the roots are 24.6° and 45.5°.

Answer
$\theta_{\text{min}} = 24.6°$

Problem 4

Given / Find: Weightless bar, $d = 1\,\text{m}$, $\theta = 30°$, resting on a smooth wall at its lower end and a smooth roller, with $P = 600\,\text{N}$ at the top. Find N_w, N_r, and a.

Sketch:



Equations & Solution: N_w is horizontal and N_r is perpendicular to the bar. Let s be the distance along the bar from the wall to the roller.

$$\sum F_y = 0: \quad N_r \cos 30° = 600 \;\Rightarrow\; N_r = 692.8\,\text{N}$$

$$\sum F_x = 0: \quad N_w = N_r \sin 30° = 346.4\,\text{N}$$

$$\sum M_{\text{wall}} = 0: \quad N_r\, s - 600(1)\cos 30° = 0 \;\Rightarrow\; s = 0.750\,\text{m}$$

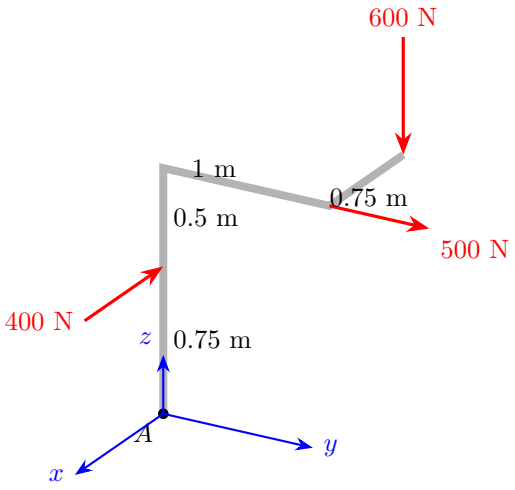
$$a = s \cos 30° = 0.6495\,\text{m}$$

Answer
$N_w = 346\,\text{N}, \quad N_r = 693\,\text{N}, \quad a = 0.650\,\text{m}$

Problem 5

Given / Find: Pipe assembly fixed at A with 400 N (along $-x$), 500 N (along $+y$), and 600 N (down) applied as shown. Find the three force and three moment reactions at A.

Sketch:



Equations & Solution: Positions from A (m) and applied forces (N):

$$\mathbf{r}_1 = \langle 0, 0, 0.75 \rangle, \qquad \mathbf{F}_1 = \langle -400, 0, 0 \rangle$$
$$\mathbf{r}_2 = \langle 0, 1, 1.25 \rangle, \qquad \mathbf{F}_2 = \langle 0, 500, 0 \rangle$$
$$\mathbf{r}_3 = \langle -0.75, 1, 1.25 \rangle, \qquad \mathbf{F}_3 = \langle 0, 0, -600 \rangle$$

Forces: $\mathbf{A} = -\sum \mathbf{F} = \langle 400, -500, 600 \rangle$ N.

Moments: $\mathbf{M}_A = -\sum \mathbf{r} \times \mathbf{F}$, with

$$\mathbf{r}_1 \times \mathbf{F}_1 = \langle 0, -300, 0 \rangle, \quad \mathbf{r}_2 \times \mathbf{F}_2 = \langle -625, 0, 0 \rangle, \quad \mathbf{r}_3 \times \mathbf{F}_3 = \langle -600, -450, 0 \rangle \text{ N.m}$$

so $\mathbf{M}_A = \langle 1225, 750, 0 \rangle$ N.m.

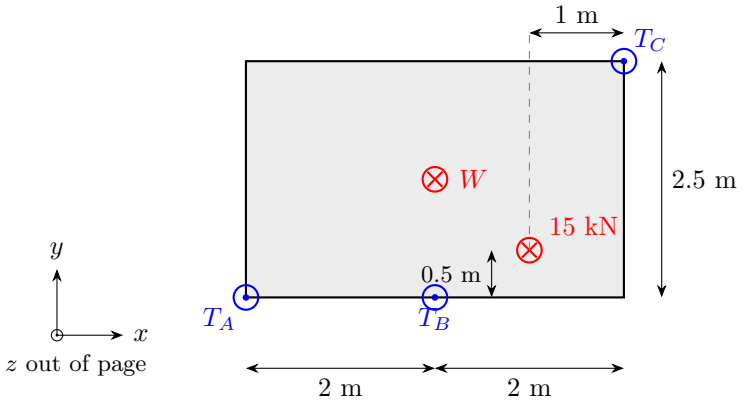
Answer

$A_x = 400\,\text{N}, \quad A_y = -500\,\text{N}, \quad A_z = 600\,\text{N}$
$M_x = 1.23\,\text{kN m}, \quad M_y = 750\,\text{N m}, \quad M_z = 0$

Problem 6

Given / Find: A 2400 kg uniform slab hangs from three vertical cables at A, B, C with an additional 15 kN load on it. Find T_A, T_B, T_C.

Sketch:



Equations & Solution: Dimensions read from the figure: slab 4×2.5 m with origin at A. $A(0,0)$, $B(2,0)$, $C(4,2.5)$, 15 kN load at $(3, 0.5)$, centroid at $(2, 1.25)$ m. $W = (2400)(9.81) = 23.54\,\text{kN}$.

$$\sum F_z = 0: \quad T_A + T_B + T_C = 23.54 + 15 = 38.54$$

$$\sum M_x = 0: \quad 2.5\,T_C = 23.54(1.25) + 15(0.5) \;\Rightarrow\; T_C = 14.77\,\text{kN}$$

$$\sum M_y = 0: \quad 2T_B + 4T_C = 23.54(2) + 15(3) \;\Rightarrow\; T_B = 16.50\,\text{kN}$$

Then $T_A = 38.54 - 16.50 - 14.77 = 7.27\,\text{kN}$.

Answer
$T_A = 7.27\,\text{kN}, \quad T_B = 16.5\,\text{kN}, \quad T_C = 14.8\,\text{kN}$